

3. The table shows the electronic structures of atoms of the elements **A–F**.

A–F are not the chemical symbols for the elements.

Element	Electronic structure
A	2
B	2,6
C	2,8,1
D	2,8,7
E	2,8
F	2,8,6

(a) Which of elements **A–F** is found in Group 6 and Period 3 of the Periodic Table?

Explain your choice, referring to electronic structure.

[2]

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(b) Which **two** of elements **A–F** are chemically inert?

Explain your choice, referring to electronic structure.

[2]

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(c) Element **D** has two isotopes. Isotope **1** has 18 neutrons and isotope **2** has 20 neutrons.

Complete the table by giving the atomic number and mass number of both isotopes. [2]

Isotope	Atomic number	Mass number
1		
2		

